Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic Workflows · RAG Platforms · LLM Evaluation · Production AI Systems · Classical Machine Learning · LangGraph · LangChain · Python · Observability · AWS

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PROFESSIONAL SUMMARY

AI Engineer with 8 years of production engineering experience and two shipped AI systems that directly map to this role's scope: AgentProbe - a RAG evaluation platform implementing evaluation strategies, metrics, and feedback loops to measure model effectiveness and detect hallucinations - and LeaseI - an 8-agent LangGraph agentic workflow with multi-source RAG, HITL escalation, and 93.6% precision across 1,240 compliance test cases. Both systems are production-hardened: CI-integrated reliability gates, telemetry-driven observability, and failure handling built in. At Entrata, owned end-to-end data pipelines, production reliability at 99.5% uptime (100K+ monthly active users), and built classical machine learning research pipelines at George Mason University (R^2=0.977, M.S. Data Analytics Engineering, GPA 3.87, May 2026). Mentored 5+ engineers; 300+ code reviews; technical guidance across AI and backend systems.

TECHNICAL SKILLS

AI Engineering: Agentic Workflows, RAG Platforms, LLM Evaluation, LangGraph, LangChain, LlamaIndex-compatible, LLM Orchestration, Prompt Engineering, Hallucination Detection, Bias & Failure Mitigation, HITL, Model Evaluation, Feedback Loops, ChromaDB, OpenAI API, Anthropic API, Hugging Face, MCP, Classical Machine Learning, PyTorch, scikit-learn, BERT, spaCy, NLP

Engineering & Data Pipelines: Python (8yr), SQL, FastAPI, PHP, JavaScript, TypeScript, Docker, Microservices, AWS (S3, EC2, RDS, Glue), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions), Observability, Reliability Engineering, Feature Engineering, Data Pipelines, PostgreSQL, MySQL, Oracle, PySpark, Linux, Git

Systems & Leadership: Production AI System Design, AI Engineering Standards, System Architecture, Performance & Cost Optimization, Technical Decision-Making, Mentoring, Code Reviews, Agile/Scrum, Cross-functional Collaboration, Statistical Modeling, Tableau

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Designed and built end-to-end agentic AI system (LangGraph) for mission planning - agentic workflow with conditional routing, RAG retrieval over distributed sources, and validation feedback loops enforcing compliance before delivery.
- Defined evaluation strategy for RAG retrieval quality and response groundedness; implemented observability and feedback loops for iterative improvement; mitigated hallucination risks through citation grounding.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built classical machine learning pipelines (Python, scikit-learn, PyTorch, R) on WHO and UK Road Safety datasets - R^2 = 0.977; end-to-end from feature engineering through model evaluation, failure analysis, and statistical validation across 190+ countries.
- Mentored 600+ students in Python, machine learning, and data science; designed dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned end-to-end data pipelines and production AI system reliability** - fragmented federal source data transformed into 180,000 domain records per release cycle; feature strategies and data quality validation at every stage; 99.5% uptime SLA at 100K+ monthly active users; zero compliance violations across 7 years.
- **Built observability and failure-handling infrastructure** - internal monitoring reducing triage time 40%; reliability engineering maintaining sub-10 bugs for 6+ consecutive months; production readiness reviews across all delivered systems.
- **System architecture and technical decision-making** - Docker microservices redesign (50% scalability improvement, 30% deployment reduction); RabbitMQ event-driven processing (retry, idempotency, dead-letter); Redis caching; AWS S3, EC2, RDS, Glue; 200+ SQL optimizations (30% gain).
- **Established engineering standards and mentored engineers** - 5+ engineers mentored on AI and backend development; 300+ peer code reviews; contributed to team-wide standards for reliability, testing, and AI system design.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Production RAG Evaluation & AI Observability Platform RAG · Evaluation

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

End-to-end AI system with defined evaluation strategy, metrics, and feedback loops: 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge auto-rater, CI-integrated regression gate blocking regressions before merge, multi-provider LLM orchestration for cost/performance optimization. Hallucination detection and bias mitigation built in. 1,240+ evaluation test cases - zero regressions shipped to production.

Leasel: 8-Agent Agentic Workflow & RAG Platform Agentic · RAG

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI system with classical machine learning hybrid: 8-agent LangGraph stateful workflow, multi-source RAG with feedback loops, HITL escalation when model confidence below threshold (failure handling), and deterministic logic controlling every regulated output. 93.6% precision, 87.9% F1 across 1,240 evaluation test cases; CI quality gate with telemetry-driven scoring.

AI on FHIR: NLP-Driven Medical Records Query System NLP · Classical ML

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

AI system combining classical NLP (BERT embeddings, spaCy) with LLM orchestration to translate natural-language queries into FHIR API calls - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017